

DATABASE MANAGEMENT SYSTEMS (CD343AI)

LAB SEMESTER END EXAMINATION

TOPIC: INTEGRATED STUDENT ACADEMIC, COUNSELLING AND CO-CURRICULAR TRACKING SYSTEM

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Introduction

- Modern educational institutions handle academic records digitally, but counseling and co-curricular activities are still largely tracked using manual or semi-digital methods.
- Studies show that fragmented data systems increase administrative workload and reduce the effectiveness of student monitoring and support.
- The absence of a unified platform results in inconsistent activity tracking, limited transparency, and inefficient counseling processes.
- **Student360** addresses this gap by integrating academics, counseling, and co-curricular management into a single, automated, student-centric system.
- This project is aligned with SDG 4: Quality Education



Problem Statement

Most educational institutions rely on separate systems to manage academics, counseling, and co-curricular activities, with counseling and activity tracking often maintained manually or through disconnected tools. This fragmented approach results in inconsistent data, increased administrative effort, and limited visibility into a student's overall development. The lack of a centralized platform also leads to inefficiencies in counselor assignment, activity point verification, and record maintenance. Consequently, institutions face challenges in providing transparent, accountable, and data-driven student support, highlighting the need for an integrated and automated management system.

Objectives

Primary Objectives

- To design a unified platform that integrates academic records, counseling processes, and co-curricular activity management.
- To automate counseling requests, club activities, and activity-point allocation to improve transparency and efficiency.

Secondary Objectives

- To provide a student-centric system that offers a holistic view of academic and extracurricular progress.
- To implement role-based access control for secure and authorized data management.
- To ensure data integrity and consistency through a normalized database design.
- To enable secure counselor–student interaction and support AI-based syllabus assistance.

Literature Review

Paper (Year)	Core Idea	Method / Model	Key Contribution	Limitations
Ali et al. (2022)	Secure student information management	Blockchain-based student information system	Improves data integrity and tamper resistance in student records	High computational cost; limited focus on counseling and co-curricular data
Majjate et al. (2023)	Personalized academic guidance	AI-based student profiling and recommendation system	Provides personalized academic and career guidance	Requires high-quality student data; limited administrative integration
Wang (2025)	Academic performance monitoring	AI-driven analytics for student performance	Enhances early intervention through intelligent monitoring	Focused mainly on academics, ignores co-curricular aspects
Yuan (2025)	Intelligent student management	Multi-modal data analysis	Integrates multiple data sources for student assessment	Increased system complexity; implementation overhead
Huang & Li (2024)	Integrated student management platform	Programmable device-based system design	Demonstrates system-level integration of student data	Hardware-dependent; lacks flexible role-based access
Mircea et al. (2021)	IoT in higher education	IoT-based educational infrastructure	Highlights potential of IoT for real-time monitoring	Infrastructure cost and scalability concerns
Jin (2025)	Special needs student management	Decision tree + big data analytics	Improves decision-making for special needs education	Highly specialized; not generalized for all students
Gunawan & Wijaya (2023)	Administrative automation	Robotic Process Automation (RPA)	Reduces manual workload in attendance and reporting	Limited to administrative tasks; no academic integration

Requirement Specification

Functional Requirements

- The system shall authenticate users through role-specific login mechanisms.
- The system shall allow students to join clubs, participate in events, and submit requests.
- The system shall enable faculty to review, approve, or reject student requests.
- The system shall allow club coordinators to manage members, events, and announcements.
- The system shall support secure messaging between students and assigned faculty.
- The system shall implement an admin-defined policy engine to detect rule violations.

Non-Functional Requirements

- Performance: Responses within 2–3 seconds
- Scalability: Supports growing users and data
- Security: Encrypted communication and access control
- Reliability: 99% system availability
- Maintainability: Modular design with minimal downtime

Hardware Requirements

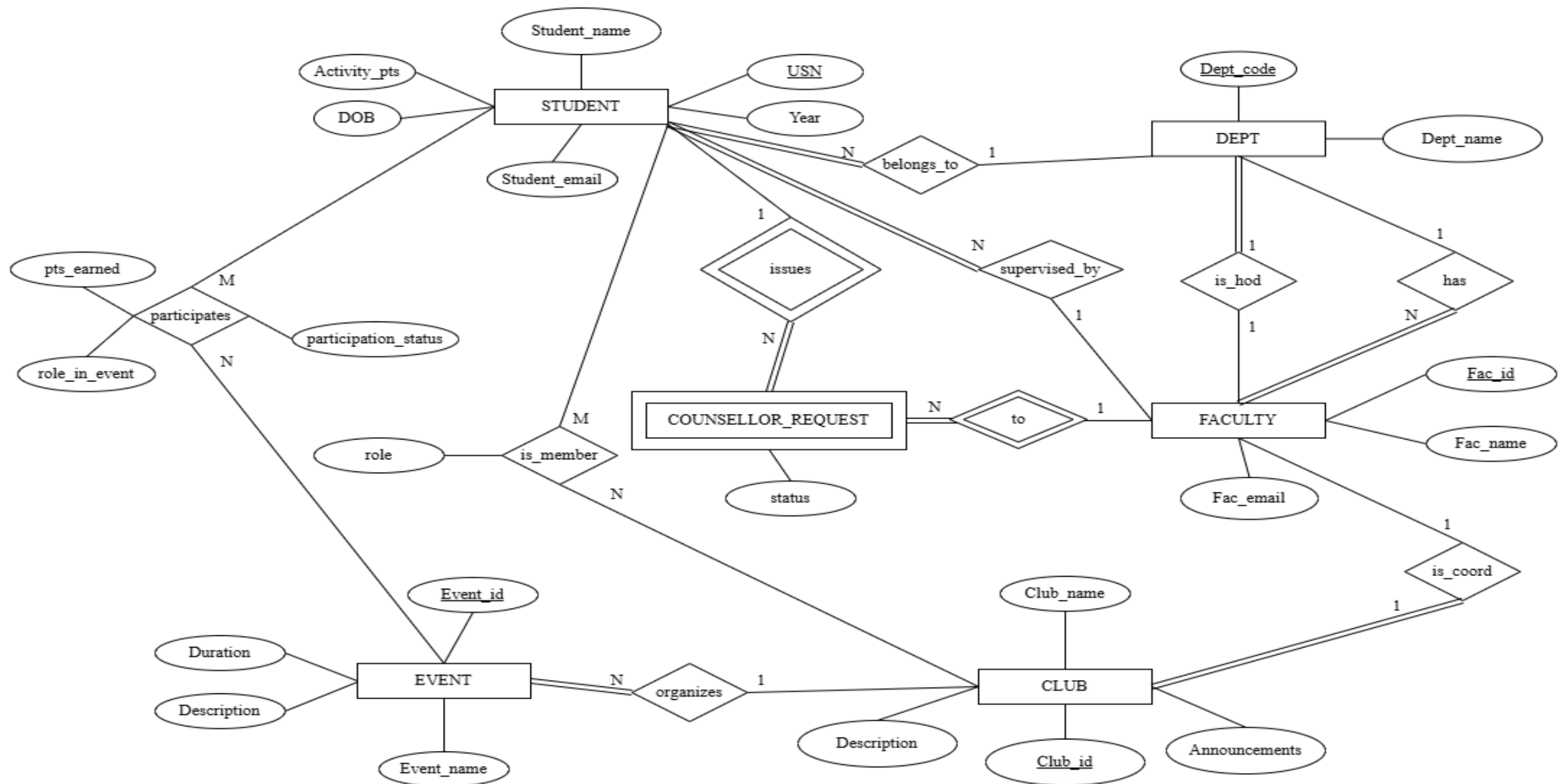
- Server: Quad-Core Intel i5 or higher, 8 GB RAM, 256 GB SSD
- Client: Dual-Core processor, 4 GB RAM
- Network: Stable high-speed internet
- Browser: Latest Chrome / Firefox / Edge

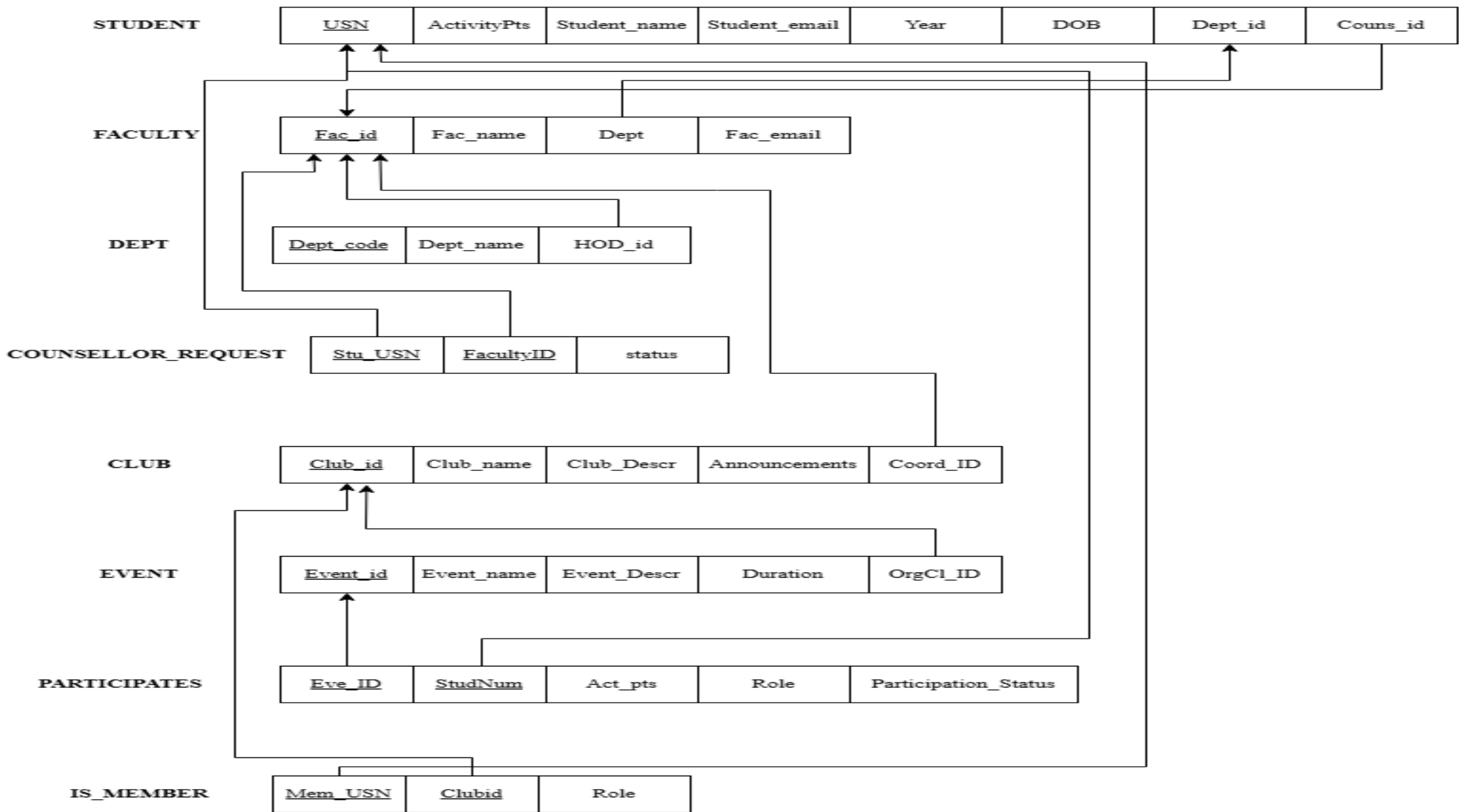
Software Requirements

- OS: Windows / Linux / macOS
- Frontend: React.js
- Backend: Node.js (Express)
- Database: MySQL
- Security: JWT authentication, RBAC
- AI Support: Syllabus assistant, Student Engagement Insights

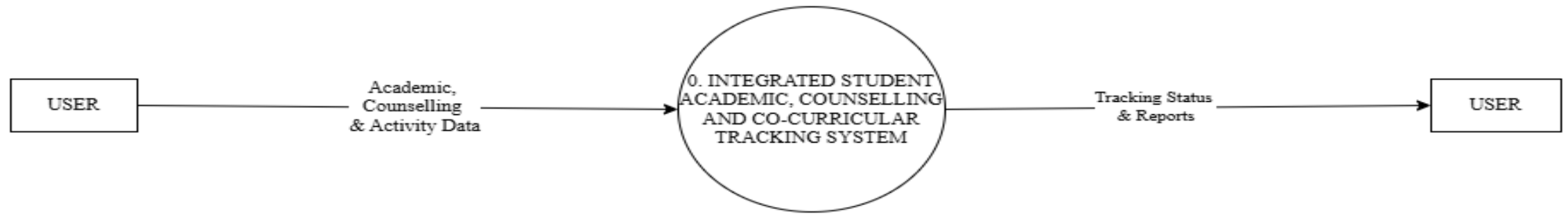
METHODOLOGY

ER DIAGRAM

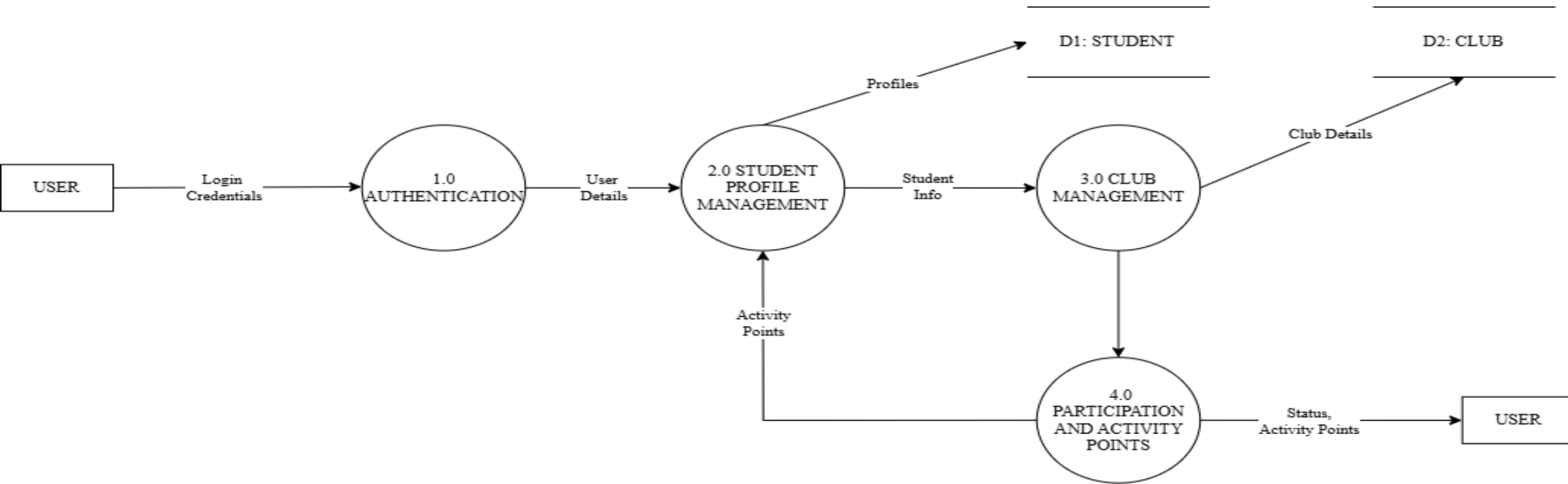




DFD LEVEL 0

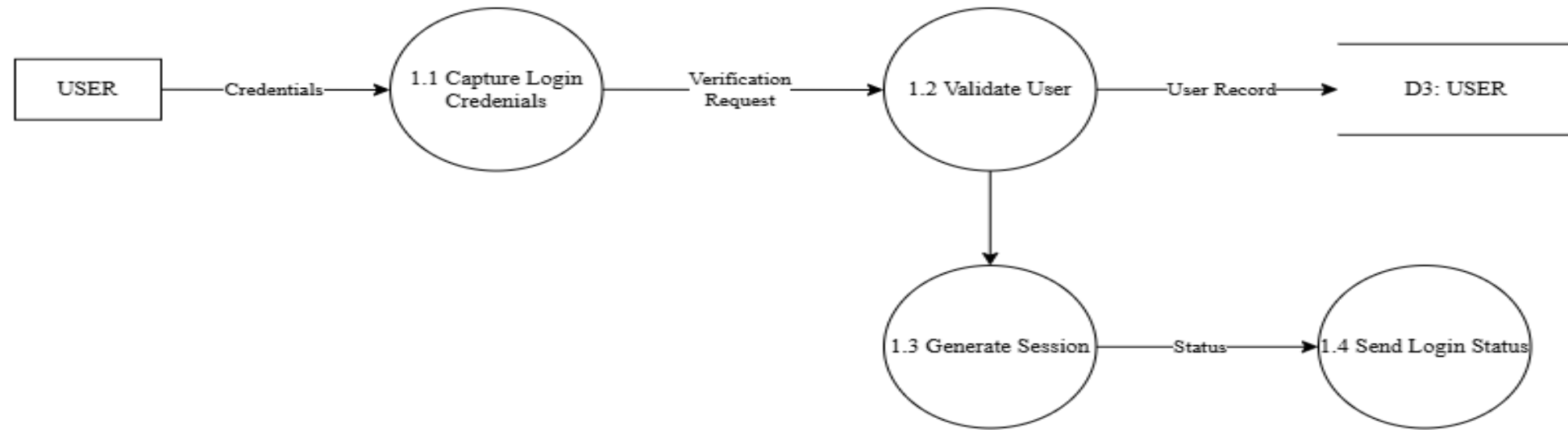


DFD LEVEL 1

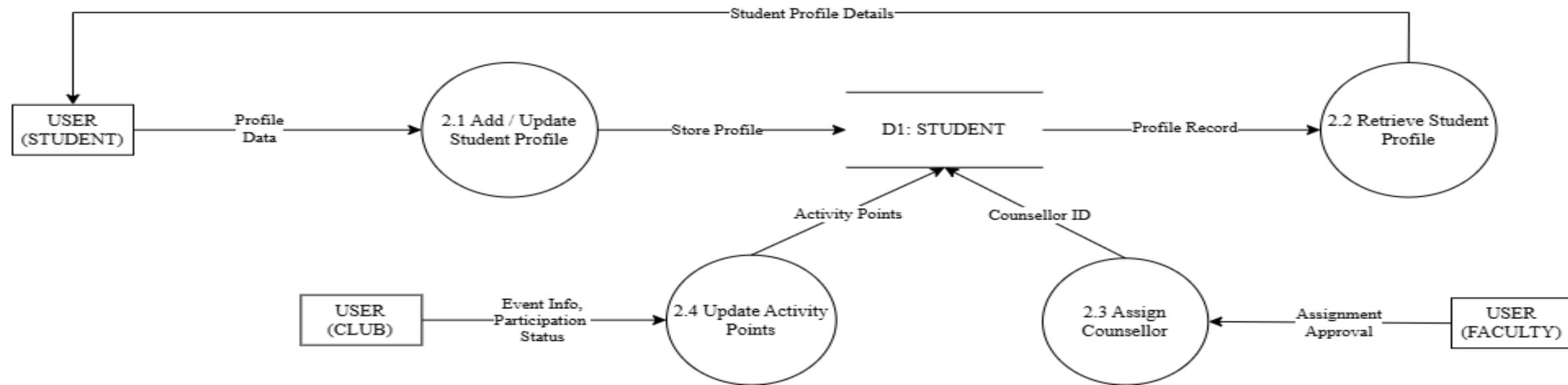


DFD LEVEL 2

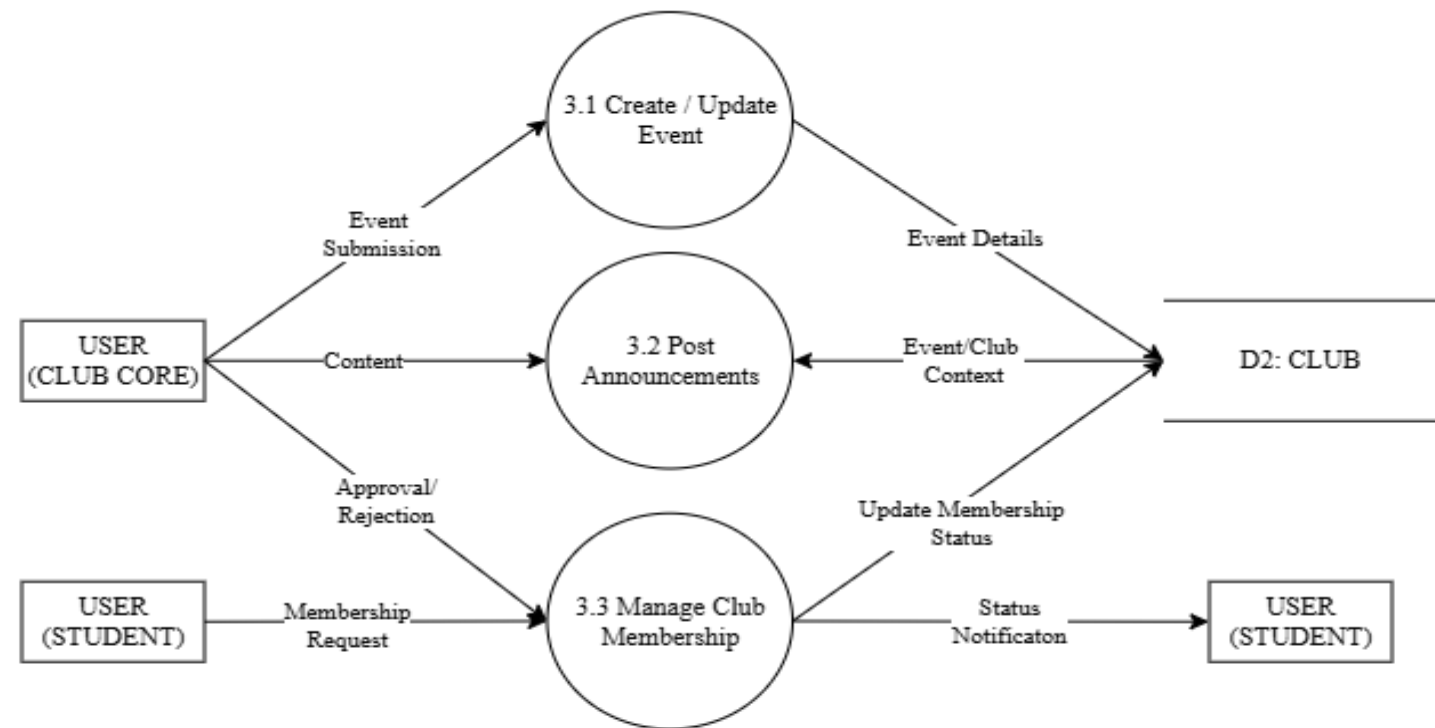
Authentication



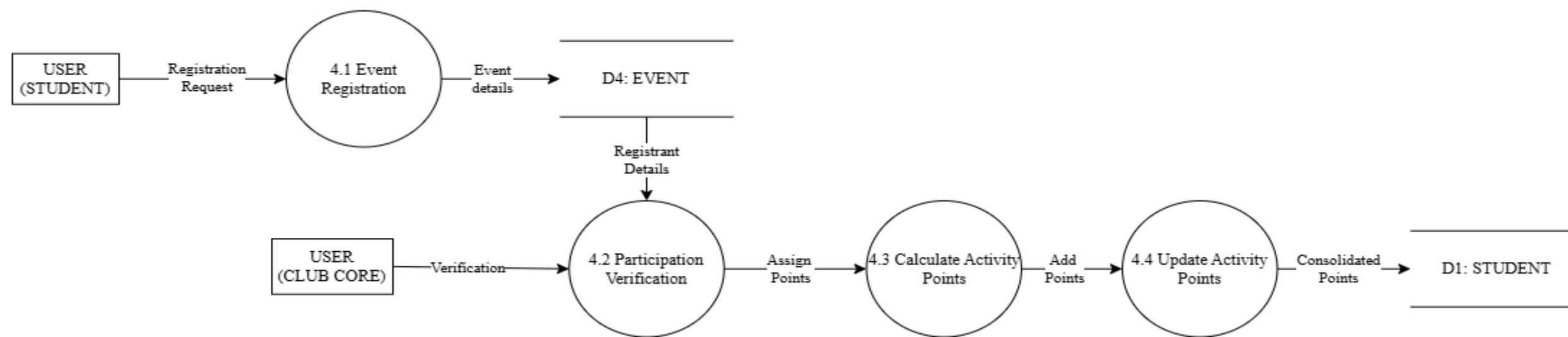
Student Profile Management



Club Management



Participation & Activity Points



Normalization

The Student360 database is **normalized up to Third Normal Form (3NF)** to ensure data integrity, eliminate redundancy, and avoid update anomalies.

- 1NF:** All attributes store atomic values with no multivalued or composite fields.
- 2NF:** All non-key attributes are fully dependent on their respective primary or composite keys, ensuring no partial dependency.
- 3NF:** No non-key attribute depends on another non-key attribute.

STUDENT
USN, Student_name, Student_email, DOB, Year, Activity_pts
USN → Student_name, Student_email, DOB, Year, Activity_pts

DEPT
Dept_code, Dept_name
Dept_code → Dept_name

FACULTY
Fac_id, Fac_name, Fac_email
Fac_id → Fac_name, Fac_email

CLUB
Club_id, Club_name, Description, Announcements
Club_id → Club_name, Description, Announcements

EVENT
Event_id, Event_name, Description, Duration
Event_id → Event_name, Description, Duration

COUNSELLOR_REQUEST
USN, Fac_id, Status
(USN, Fac_id) → Status

PARTICIPATES
USN, Event_id, Role_in_event, Participation_status, Pts_earned
(USN, Event_id) → Role_in_event, Participation_status, Pts_earned

IS_MEMBER
USN, Club_id, Role
(USN, Club_id) → Role

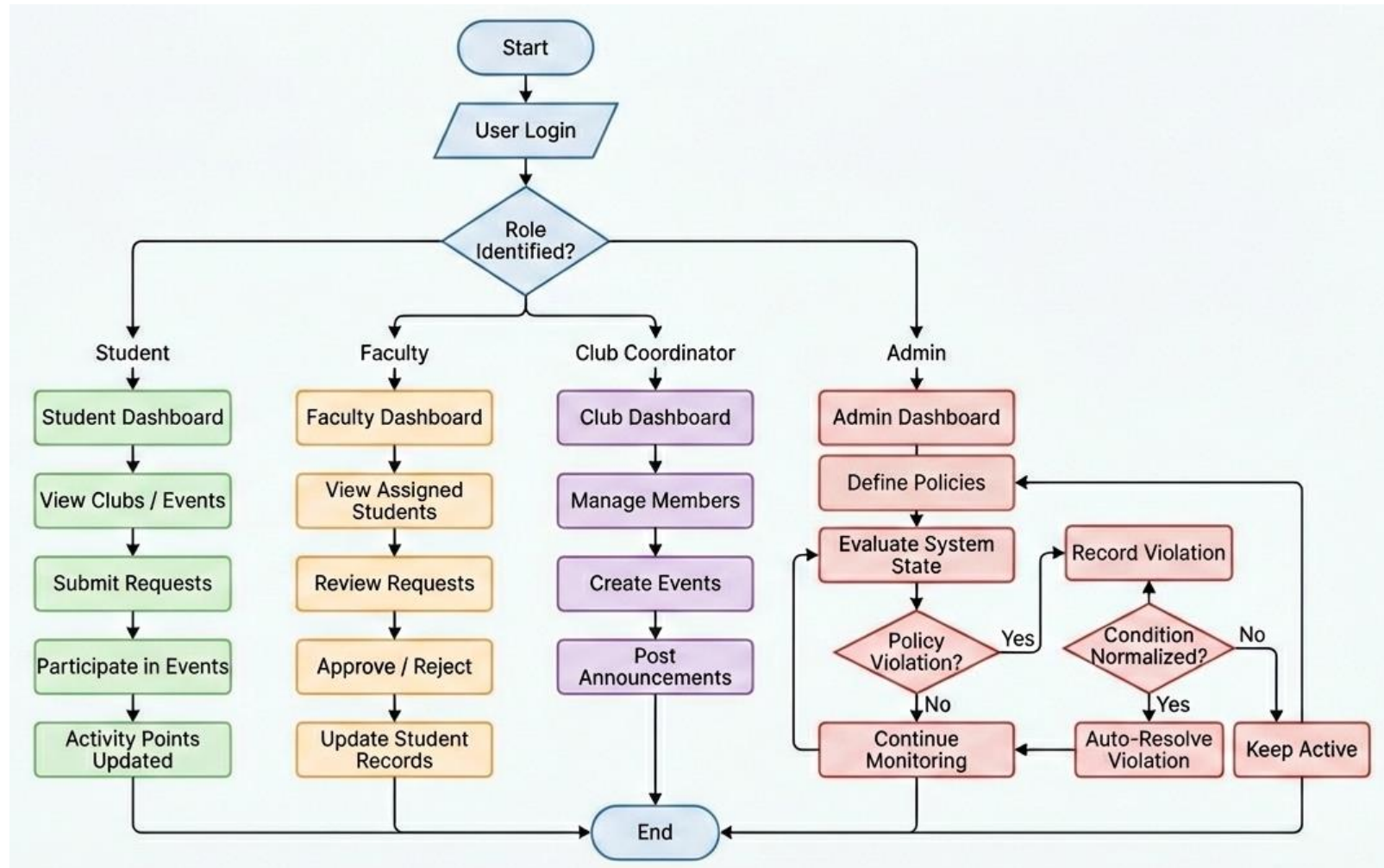
Methodology

System Design Approach:

- The system follows a **modular and layered architecture** integrating academic, counseling, and co-curricular modules.
- Core entities (Admin, Student, Faculty, Club, Event) are stored in a relational database, while dynamic insights are handled separately.

Methodology Steps:

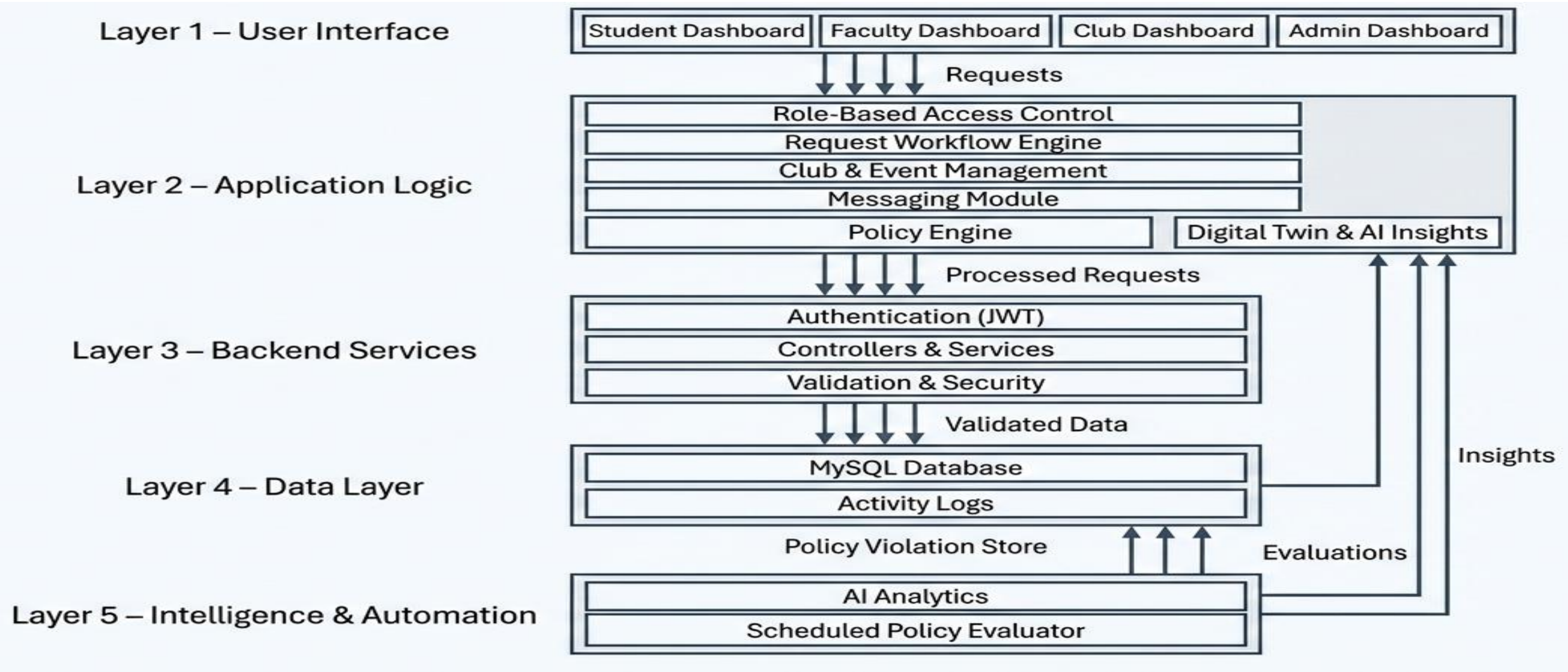
- Users enter the system through role-specific login (Student / Faculty / Club / Admin)
- Each role is routed to a dedicated dashboard showing only relevant actions and data



Methodology

- Students interact with the system through:
 - **Club discovery → join requests → approval → club feed access**
 - **Event browsing → registration → participation → activity point updates**
 - **Structured requests (counsellor join, activity points) routed to faculty**
 - **All actions follow a request → validation → approval/rejection → state update flow**
- Faculty act as **academic supervisors**, reviewing requests, monitoring students, and updating records
- Club coordinators manage **members, events, and announcements**, influencing student activity data
- Admin continuously governs the system using a **policy-driven evaluation cycle**:
 - Monitor system state
 - Detect rule violations
 - Record violations
- Student data feeds into **Digital Twin & AI insights**, creating a **feedback loop** for early intervention
- Every transition is logged to ensure **traceability, accountability, and audit readiness**

Architecture



Role of DBMS – Query-Driven System Functionality

- **JOIN operations (INNER, LEFT JOIN)** are used to combine related data across tables, enabling features like student–club membership status and counsellor–student mapping.
- **LEFT JOIN** queries ensure complete data visibility, such as displaying all students or clubs even when related records (membership, requests) are missing.
- **Aggregate functions (COUNT, SUM)** support real-time analytics, including counsellor workload calculation and activity point totals.
- **Conditional queries (WHERE, HAVING)** enforce business rules such as eligibility checks and policy threshold evaluation.
- **Transaction-safe updates** maintain consistency during approvals, rejections, and activity point modifications.
- **Indexing and normalized tables** enable fast retrieval and scalable performance across dashboards and reports.

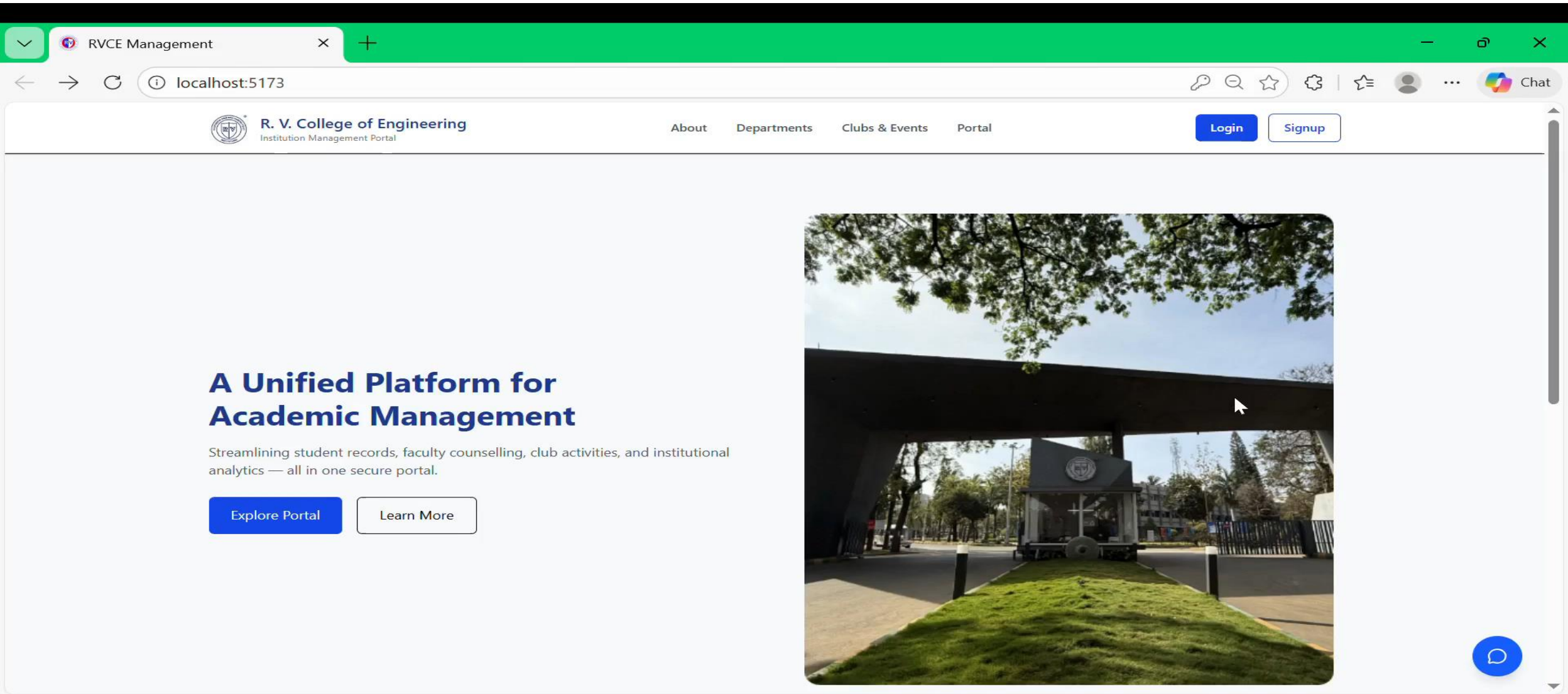
Components and Tools Used

Layer	Component	Description
User Interface	Student Dashboard	Enables students to access profile, clubs, events, requests, messages, and activity points
	Faculty Dashboard	Allows faculty to manage students, requests, counselling, and view insights
	Club Dashboard	Used by coordinators to manage members, events, and announcements
	Admin Dashboard	Used to configure policies and monitor system-wide status
Application Logic	Role-Based Access Control	Restricts system features based on authenticated user roles
	Request Workflow Engine	Manages request lifecycle: submission, validation, approval, rejection
	Club & Event Management	Handles club membership, event creation, participation, and attendance
	Messaging Module	Supports secure student–faculty communication
	Policy Engine	Evaluates institutional rules and identifies violations
	Digital Twin & AI Insights	Analyzes student behavior, engagement, and risk indicators

Components and Tools Used

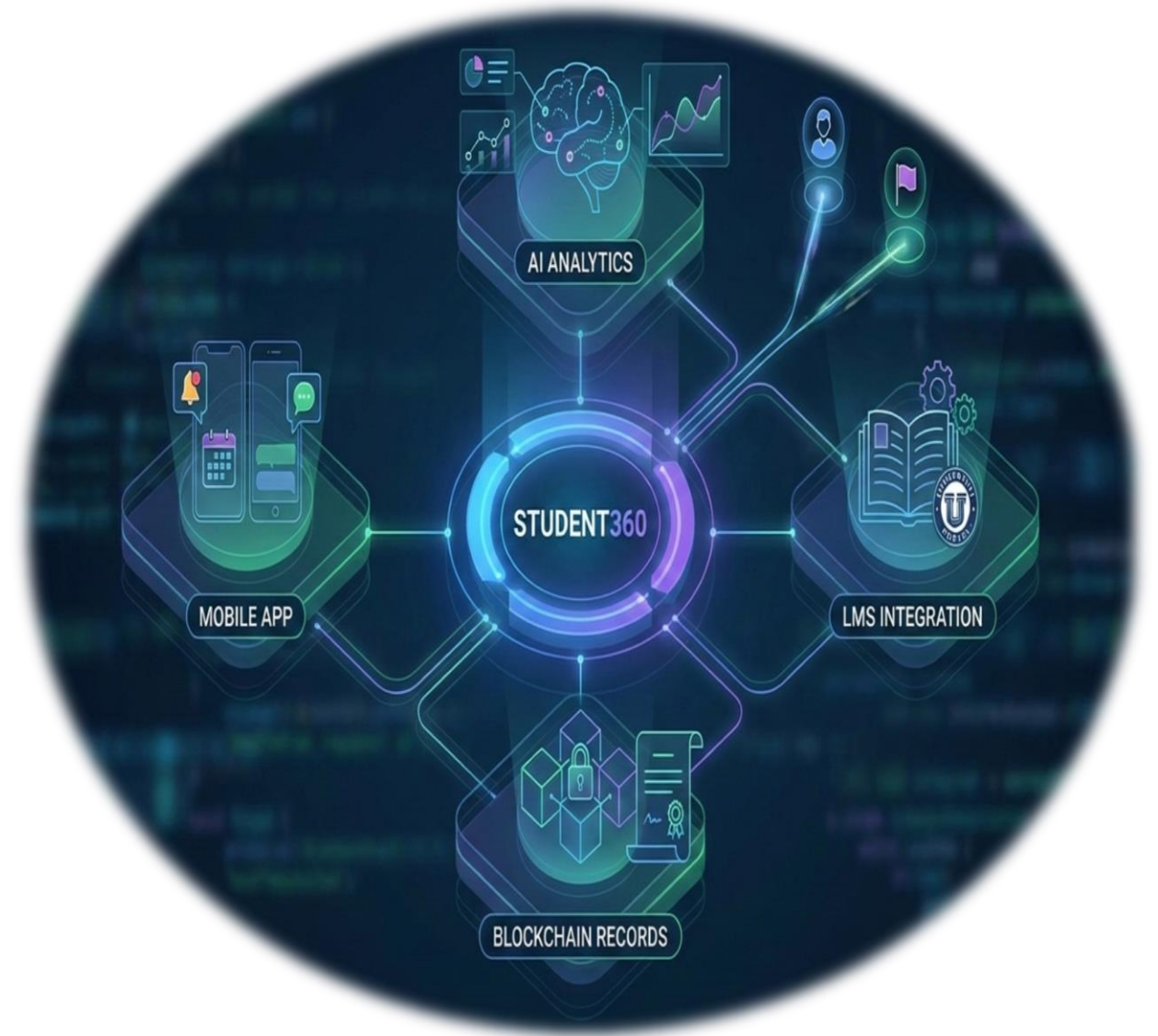
Layer	Component	Description
Backend Services	Authentication Service (JWT)	Authenticates users and issues secure access tokens
	Controllers & Services	Implements core business logic and orchestrates workflows
	Validation & Security	Enforces server-side checks and prevents unauthorized access
Data Layer	MySQL Database	Stores all institutional data with relational integrity
	Activity Logs	Maintains audit trails for actions and system events
	Policy Violation Store	Records detected and resolved policy violations
Intelligence & Automation	AI Analytics	Generates predictive and behavioral insights from data
	Scheduled Policy Evaluator	Periodically checks policies and auto-resolves violations
	AI-based Syllabus Assistant	Helps students to gather information about the syllabus

Demo Video



Future Scope

- **Advanced AI Integration:** Incorporate predictive analytics to identify students at academic or counseling risk and recommend early interventions.
- **Mobile Application Support:** Extend the platform to Android and iOS applications for real-time notifications, event updates, and counseling reminders.
- **Blockchain-Based Records:** Use blockchain technology to securely store academic credentials and activity certificates with tamper-proof verification.
- **Integration with Learning Management Systems (LMS):** Connect Student360 with existing LMS platforms to analyze academic performance alongside co-curricular involvement.



Conclusion

- The **Student360 system** provides a unified, student-centric platform integrating academics, counselling, club activities, and institutional governance.
- The project replaces manual and fragmented workflows with **structured, role-based digital processes**, improving efficiency and transparency.
- A **well-normalized relational database design (3NF)** ensures data integrity, scalability, and accurate modeling of real-world institutional relationships.
- The inclusion of a **policy-driven governance engine** enables automated violation detection and resolution, strengthening administrative control.
- Secure role-based access control** ensures data privacy, accountability, and system reliability.
- Overall, the system demonstrates a scalable and future-ready solution, with scope for AI-driven analytics, mobile integration, and advanced reporting.

References

- [1] S. I. M. Ali, H. Farouk, and H. Sharaf, "A blockchain-based models for student information systems," *Egyptian Informatics Journal*, vol. 23, no. 2, pp. 187–196, Jul. 2022, doi: [10.1016/j.eij.2021.12.002](https://doi.org/10.1016/j.eij.2021.12.002).
- [2] H. Majjate, Y. Bellarhmouch, A. Jeghal, A. Yahyaouy, H. Tairi, and K. A. Zidani, "AI-Powered Academic Guidance and Counseling System Based on Student Profile and Interests," *ASI*, vol. 7, no. 1, p. 6, Dec. 2023, doi: [10.3390/asi7010006](https://doi.org/10.3390/asi7010006).
- [3] Y. Wang, "Artificial intelligence in student management systems to enhance academic performance monitoring and intervention," *Sci Rep*, vol. 15, no. 1, p. 35122, Oct. 2025, doi: [10.1038/s41598-025-19159-4](https://doi.org/10.1038/s41598-025-19159-4).
- [4] G. Yuan, "Design and evaluation of intelligent student management system based on multi-modal data," *Systems and Soft Computing*, vol. 7, p. 200370, Dec. 2025, doi: [10.1016/j.sasc.2025.200370](https://doi.org/10.1016/j.sasc.2025.200370).
- [5] H. Huang and B. Li, "Design and implementation of student management system of integrated programmable device programming system," *Sci Rep*, vol. 14, no. 1, p. 11873, May 2024, doi: [10.1038/s41598-024-62844-z](https://doi.org/10.1038/s41598-024-62844-z).
- [6] M. Mircea, M. Stoica, and B. Ghilic-Micu, "Investigating the Impact of the Internet of Things in Higher Education Environment," *IEEE Access*, vol. 9, pp. 33396–33409, 2021, doi: [10.1109/ACCESS.2021.3060964](https://doi.org/10.1109/ACCESS.2021.3060964).
- [7] J. Jin, "Research and construction of student management platform for special needs students with decision tree model and big data technology," *Systems and Soft Computing*, vol. 7, p. 200310, Dec. 2025, doi: [10.1016/j.sasc.2025.200310](https://doi.org/10.1016/j.sasc.2025.200310).
- [8] A. Gunawan and M. I. Wijaya, "Robotic Process Automation to Enhance Education's Administration Process: Case of Attendance Checking and Reporting," in *2023 International Conference on Information Management and Technology (ICIMTech)*, Malang, Indonesia: IEEE, Aug. 2023, pp. 1–6. doi: [10.1109/ICIMTech59029.2023.10278035](https://doi.org/10.1109/ICIMTech59029.2023.10278035).
- [9] Krithika. D. R and S. V M, "Smart Event Management System Using Machine Learning for Optimized Scheduling and Participant Behavior Tracking," in *2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE)*, Virudhunagar, India: IEEE, May 2025, pp. 1–4. doi: [10.1109/ICCRTEE64519.2025.11052976](https://doi.org/10.1109/ICCRTEE64519.2025.11052976).
- [10] V. Bhardwaj and M. Kumar, "Transforming higher education with robotic process automation: enhancing efficiency, innovation, and student-centered learning," *Discov Sustain*, vol. 6, no. 1, p. 356, May 2025, doi: [10.1007/s43621-025-01198-6](https://doi.org/10.1007/s43621-025-01198-6).

Thank You!